MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

Yunheng Zou[2,6,8], **Olalla Nieto Faza**[10], **Shifa Hussain**[11], **Varinia Bernales**[2,5,†], **Alán Aspuru-Guzik**[1,2,3,4,5,6,7,8,9,†]

[1]Department of Chemistry, University of Toronto, 80 St. George St., Toronto, ON M5S 3H6, Canada
[2]Department of Computer Science, University of Toronto, 40 St George St., Toronto, ON M5S 2E4, Canada
[3]Department of Materials Science & Engineering, University of Toronto, 184 College St., Toronto, ON M5S 3E4, Canada
[4]Department of Chemical Engineering & Applied Chemistry, University of Toronto, 200 College St., Toronto, ON M5S 3E5, Canada
[5]Acceleration Consortium, 700 University Ave., Toronto, ON M7A 2S4, Canada
[6]Vector Institute for Artificial Intelligence, W1140-108 College St., Schwartz Reisman Innovation Campus, Toronto, ON M5G 0C6, Canada
[7]Canadian Institute for Advanced Research (CIFAR), 661 University Ave., Toronto, ON M5G 1M1, Canada
[8]NVIDIA, 431 King St. W #6th, Toronto, ON M5V 1K4, Canada
[9]Institute of Medical Science, 1 King's College Circle, Medical Sciences Building, Room 2374, Toronto, ON M5S 1A8, Canada
[10]Department of Organic Chemistry, Universidade de Vigo, Spain
[11]Department of Biology, University of Toronto Mississauga, 1867 Inner Circle Rd., Mississauga, ON L5L 1C6, Canada
[†]Principal investigators

Atom-to-atom mapping traces atom identities from reactants to products, supporting reaction analysis and pathway construction. Yet the same endpoints can admit multiple correspondences, and selecting a single answer can conceal alternatives with different chemical implications. We introduce MAPPA, a framework that efficiently discovers and retains alternative atom mappings. It uses shared molecular fragments and molecular symmetry to represent these alternatives compactly, allowing their bond changes to be examined and supporting subsequent chemistry-aware selection. MAPPA achieves the highest reference coverage among the evaluated methods on Golden, reaching 99.46%, and recovers 99.63% of EnzymeMap reference mappings. In the standard benchmark runs, mapping and decoding together average 3.19 CPU seconds per Golden reaction and 13.16 CPU seconds per EnzymeMap reaction. These results make broad exploration of mapping alternatives practical, providing a foundation for studying reaction ambiguity and incorporating chemical knowledge into mapping selection.

Date: November 8, 2026
Status: Preprint

1 Introduction

A chemical reaction connects two endpoints: the reactant structures (R) and the product structures (P), each of which may contain several molecules. Given these endpoints, atom-to-atom mapping (AAM) assigns corresponding atoms between R and P, preserving their elements and using each mapped atom only once. This correspondence identifies which reactant atom becomes which product atom. Comparing the bonds between corresponding atom pairs then reveals the net bond changes: for example, a bond present in R but absent in P is a cleavage under that mapping. AAM therefore describes the endpoint result of chemical evolution; the intervening sequence of elementary steps requires additional information.

These assignments make reactions usable as structured data. They locate reaction centers and enable the extraction of transformation templates for reaction prediction and retrosynthesis (14; 1). They also establish atom identities between endpoint geometries for reaction-path calculations (3). A mistaken correspondence can change the inferred transformation and propagate into downstream analyses. MAPPA addresses this task by exploring shared molecular fragments and the alternative mappings they support (Figure 1).

The difficulty is that endpoint structures alone may not select a unique correspondence. Repeated groups offer several possible matching sites, and molecular symmetry permits exchanges that leave a structure unchanged. Other assignments can imply different atom origins or different bond changes. For example, when several reactant molecules contain oxygen, deciding which oxygen appears in a product may require

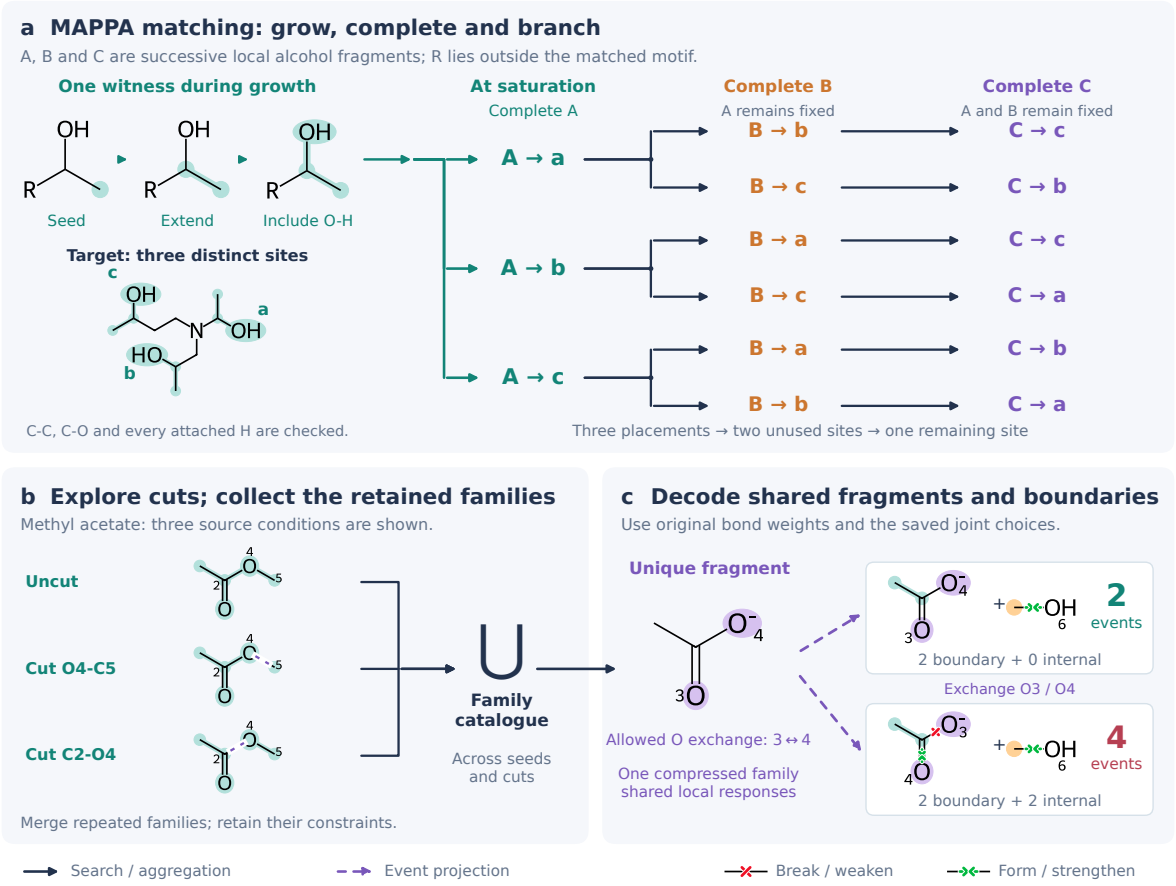


Figure 1 MAPPA discovers and organizes alternative atom mappings. (a) Growth retains one witness; saturation constructs compressed placements for conditional branching. (b) Seeds and single-edge cuts contribute families to a shared catalogue. (c) Decoding reuses unique-fragment symmetry responses and combines internal, boundary and sweep-bond contributions. Red crosses mark cleavage or weakening; green arrows mark formation or strengthening. Dashed arrows project choices within a compressed family. Molecular examples and their scope are explained in Appendix A.

chemical knowledge beyond element identity. Favoring a small number of bond changes is a useful structural preference, but does not by itself establish the chemically correct atom history (1). A mapper must therefore balance the cost of exploring alternatives with the risk of discarding a useful one.

Existing approaches use structural optimization or information learned from reaction data to guide these assignments. Structural methods seek shared substructures or favorable bond-change patterns (10); SLAPMapper uses iterative atom-label refinement and assignment optimization (9). RXNMapper obtains correspondence information from a reaction language model (14), while LocalMapper learns from chemist-labeled examples (1). For the present work, a further distinction is useful: whether the intended output is a set of alternatives or one selected prediction (Table 1). This distinction concerns the answer supplied to the user and can apply to either structural or learned methods.

Table 1 Two output objectives for atom-to-atom mapping. The distinction separates discovery of alternatives from selection of a preferred answer. The examples illustrate these roles rather than an exhaustive classification of AAM methods.

	Generate alternatives	**Predict one mapping**
Goal	Retain different correspondences allowed by the search rules.	Select the correspondence judged most plausible.
Output	A candidate set or a compact family of mappings.	One selected atom assignment.
Downstream use	Examine ambiguity or apply a separate chemical selector.	Use the selected assignment directly.
Evaluation	Coverage: the reference is among the returned alternatives.	Top-1 accuracy: the selected answer matches the reference.
Examples	SLAPMapper (9); MAPPA (this work).	RXNMapper (14); LocalMapper (1).

The two objectives can be combined in a discovery-and-selection workflow. An alternative-generating solver supplies candidates; a selector then chooses among them using a preference such as fewest bond events, chemical rules or a learned ranking. Coverage measures whether the reference remains available for that later choice, whereas top-1 accuracy measures whether the choice is correct. High coverage is therefore useful but does not establish equally high top-1 accuracy. MAPPA focuses on the discovery step and retains alternatives for subsequent interpretation and selection.

The central challenge for this objective is to preserve useful diversity without repeatedly processing equivalent atom assignments. MAPPA treats mapping as a constraint-search problem: shared molecular fragments restrict which atoms can correspond, while compact symmetry relationships retain allowed rearrangements. Matching discovers these fragments and their combinations; reduction collects repeated fragments and branch compositions; decoding relates the retained alternatives to bond changes. This separation allows fragment calculations to be reused across alternatives and supports parallel exploration. We evaluate reference coverage, computational cost and the choices remaining under a minimum-bond-event preference on Golden (10) and EnzymeMap (7).

2 Methods

2.1 Overview: from shared fragments to alternative mappings

MAPPA builds atom mappings by growing structural fragments shared between reactants and products (Figure 1). Starting from a seed atom, it extends a fragment while a compatible placement remains possible. When no admissible extension remains, it finds the completed fragment's alternative placements. Each placement defines a search branch: that choice is locked, and fragment growth continues on the remaining atoms under the accumulated assignments. Repeating this process produces alternative compositions of matched fragments. Different seeds, single-bond cuts and search directions expose additional compositions.

Three stages organize this procedure. The *matching stage* discovers fragments and their conditional placements. The *reduction stage* collects repeated fragments and branch compositions into a shared mapping catalogue. The *decoding stage* uses that catalogue to relate permitted atom rearrangements to bond events, preparing each unique fragment once and combining its response with the branch boundaries. Its output retains branches that can be queried within a chosen event-count window, rather than only one selected mapping. Matching and decoding are separate operations; a saved catalogue can be decoded without repeating discovery. Reference verification evaluates these outputs separately (Section 3.2).

2.2 Matching Stage: greedy fragment growth and hierarchical branching

Consider one search direction, with a source and a target molecular graph. Atoms carry element labels, and bonds carry their original weights; explicit hydrogens participate. A fragment begins at a source seed and grows through neighboring connections. Each proposed extension asks whether the enlarged fragment has at least one compatible, one-to-one placement in the target. An explicit valid assignment is called a *witness*. A positive answer accepts the extension and updates that witness. An impossible extension is deferred while other frontier connections are explored. Growth stops when no admissible extension remains.

Compatibility requires equal elements, distinct target atoms, target bond presence and a source–target bond-weight difference within the discovery tolerance τ_{iso}. The presence threshold b determines which source connections can be explored. Growth selects a boundary atom uniformly, then an incident frontier edge uniformly; weights do not rank proposals. Only accepted growth edges become preservation constraints. Other bonds between the included atoms are not automatically required, and source nonedges impose no prohibition. Consequently, the accepted growth graph can differ from the completed fragment's full internal connectivity. Coordinates may accompany the input but do not determine growth.

Finding one witness is sufficient to decide whether growth can continue. The current assignment is extended when possible; otherwise MAPPA can rearrange assignments within the growing fragment or reuse a compatible previously discovered fragment. Assignments locked by earlier fragments remain fixed. If a search is needed, it stops at the first witness; only exhausted alternatives establish absence. A resource limit is recorded as incomplete computation. Joining a previously completed fragment retains its locked assignments and accepted constraints.

At saturation, the question changes from whether a match exists to where the completed fragment can be placed. MAPPA searches the full accepted constraints for distinct target atom sets, representing their internal alternatives compactly. For example, a six-carbon ring can occupy either of two target rings with distinct surrounding environments. These are two atom-set placements; rotations within either ring do not create additional placements. The search excludes already represented placements directly, avoiding enumeration of every internal bijection followed by merging.

Each retained placement conditions the next fragment search. If fragment A occupies one target site, fragment B must use the remaining compatible sites, as illustrated in Figure 1a. Whole-endpoint symmetry can sometimes connect different atom-set placements and allow their later searches to be shared. This is used only when the transformation preserves every subsequent matching constraint and earlier lock; it moves the continuation jointly. Otherwise the placements remain separate. The branch cap limits retained continuations through pruning from the leaves upward, using the reduced descendant counts at each level. Internal symmetry permutations do not count as separate branches.

This construction treats mapping as constraint discovery. A locked fragment expresses a structure that can coexist in both endpoints, restricting the choices left for subsequent fragments. Greedy growth extends that constraint as far as the current trajectory permits. The fragment is locally saturated, not necessarily the

largest common fragment obtainable from every starting point. This motivates the exploration strategy below; precise matching conditions and hierarchy rules are given in Appendix A.

2.3 Matching Stage: exploring seeds and single-bond cuts

One greedy trajectory can repeatedly favor a large preserved region and miss another useful decomposition. In Figure 2a, growing the large B–A region first tends to leave R–B as the next fragment. If B–A contains 90% of the eligible seed atoms, a uniformly chosen initial seed falls there with probability 0.9. Additional seeds can discover different constraints, but may also revisit the dominant decomposition.

A single-edge *cut sweep* perturbs this commitment systematically. MAPPA searches the uncut graph and then repeats the search with each eligible source edge temporarily masked, including bonds to hydrogen. Each cut condition receives the requested number of seed orderings. Masking a connection inside the dominant region can expose a trajectory that ordinary seeding rarely reaches. The original endpoint weights remain unchanged for decoding: a mask alters discovery and does not assert a broken bond. Its endpoints may reconnect through another route, and one cut can redirect several later fragment choices.

Searches in both reaction directions add further alternatives. Seeds, cuts and reverse searches broaden discovery without making it exhaustive. The same exploration principle can be applied to another mapper; our expanded SLAP protocol reaches approximately 97% coverage, although its changes in both search scope and weight mode prevent a sweep-only attribution (Section 4.4).

2.4 Reduction Stage: collecting unique fragments and branch compositions

The searches above often rediscover the same fragment. Reduction identifies it by its source and target atom-set pair (R_i, P_i) and gives that pair one shared record. For example, if two seed searches both match source atoms $\{1, 2, 3\}$ to target atoms $\{a, b, c\}$, they reference the same completed fragment even if their growth order or starting witness differs. Each branch becomes a composition of these shared fragment identities, allowing repeated compositions to be removed before decoding.

The retained record keeps one original witness, its discovery constraints, anchors and symmetry information intact. Duplicate records are discarded rather than combined into broader mapping permissions. Branches also retain their ordered placement context, because a shared placement transformation must move its continuation consistently. A *mapping family* stores these retained local relations and placements with their alternative partitions and cut provenance. Different fragment partitions remain alternatives, not contributions to be added together.

Reduction reuses the fragment inventories accumulated during matching. It performs structural grouping and deduplication, without symmetry optimization or bond-event computation. Selecting one record per atom-set pair does not preserve the union of every discarded discovery relation; this restriction remains part of the method. The output is one catalogue per reaction and search direction, combining the requested seeds and cuts. That complete catalogue supplies the decoder.

2.5 Decoding Stage: local fragment symmetry and internal bond events

Decoding first identifies the unique fragments and which branches use them. Its goal is to compute a fragment's local symmetry-to-bond-event relationship once, then reuse it across all those branches. A response records which rearrangements remain possible and how they change the internal event count. Branch assembly follows in Section 2.6.

Local symmetry is computed on each fragment's own binary internal graph, preserving elements and fixing its anchors. Bonds satisfying the presence criterion share one structural label; their original weights remain available for event counting. External attachments do not define the fragment's internal symmetry. Nauty supplies compact generators (11), each describing a permitted joint rearrangement. For example, exchanging two identical arms moves each arm with its attached atoms, while an internal hydrogen exchange can act within one arm. The arm choices must still occupy different positions. These correlated operations represent the alternatives without storing every complete permutation. Both endpoint actions are retained; for an aligned equal-connectivity pair, one side suffices to describe their relative rearrangements.

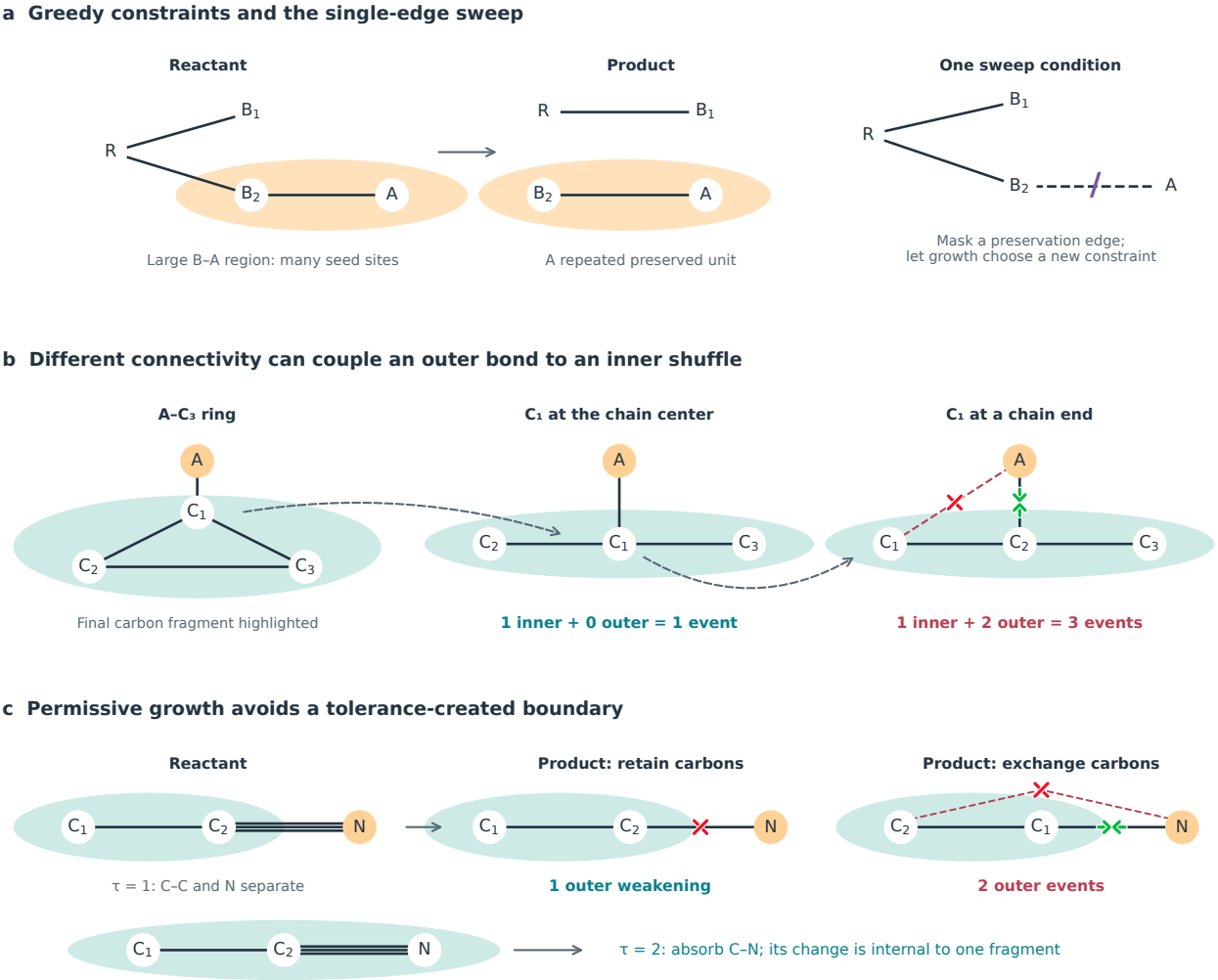


Figure 2 Structural constraints guide exploration and decoding. (a) A large preserved B–A region can dominate greedy growth; masking one of its preservation edges perturbs that commitment. R, A and B denote schematic groups, not measured reaction frequencies. (b) An A-substituted carbon triangle maps to an A-substituted carbon chain. Both displayed carbon assignments have one internal cleavage. Placing the A-bearing carbon at the chain center preserves the outer attachment; placing it at an end adds a cleavage and a formation. The physical totals are one and three, respectively. Dashed correspondence lines follow the same indexed atoms. (c) On the displayed heavy-atom graphs, tolerance one can leave C–C and N separate when a triple bond becomes single. A carbon exchange then changes outer events from one to two. Tolerance two allows the connection inside one fragment. Panels b and c are analytically checked graph examples with unit bond orders except the indicated triple bond, event threshold 0.5, and no hydrogen constraints. They explain the independence approximation and do not assert a measured failure frequency. The visual bond-event symbols follow Figure 1.

Events are evaluated from the original bond-weight differences under the resulting correspondence. A change counts when its magnitude is strictly greater than the event threshold, with its sign distinguishing formation or strengthening from cleavage or weakening. Our studies use 0.5, or 0.3 for a pair involving a metal. Discovery tolerance and event thresholds therefore serve different purposes: a permitted binary rearrangement can preserve connectivity while changing weighted bond events. In the acetate example (Figure 1c), exchanging the two oxygen images adds two internal bond events even though the binary connectivity is preserved. The response separates independent components when no event term or requested endpoint couples them, and shares computations for event-invariant choices.

For each fragment, the decoder improves the saved starting correspondence under its local symmetries. The growth graph and final internal connectivity can nevertheless differ, as in the substituted carbon triangle and chain in Figure 2b. The displayed rearrangements preserve the one-event internal ring-opening cost while changing the outer attachment. More generally, useful starting correspondences need not be connected by local symmetry. Additional starting-correspondence search is therefore restricted to fragments with such a connectivity mismatch whose current witness remains above a valid lower bound. Preparation and improvement share a fixed per-fragment budget. The retained result consists of one selected starting correspondence and its queryable symmetry response; unless a valid bound or complete search certifies optimality, its minimum is best found. Selecting this starting orbit is an explicit coverage restriction.

2.6 Decoding Stage: boundary events, sweep correction and event windows

A branch combines the prepared internal responses with events at its fragment boundaries. MAPPA aims to share local calculations rather than enumerate whole-reaction assignments. It uses the independent ordinary-boundary approximation: ordinary boundary contributions stay fixed, while the single swept connection retains its dependence on the two endpoint assignments. For a branch $\mathcal{B}$ and a permitted correspondence m, the reported count is

$$E_{\mathcal{B}}(m) = B_{\mathcal{B}} + \sum_i e_i(m) + s_{\mathcal{B}}(m), \qquad 0 \le s_{\mathcal{B}}(m) \le 2. \tag{1}$$

Here e_i counts internal events owned by a disjoint final fragment piece, $B_{\mathcal{B}}$ is the ordinary-boundary contribution, and $s_{\mathcal{B}}$ is the swept bond's correction. Bonds touching mapped atoms are counted, including mapped/unmapped boundaries; bonds wholly inside unmatched material are omitted. No correspondence is invented for an unmatched atom. Assembly reuses the local responses and preserves shared choices, one-to-one occupancy and the correlation between the sweep endpoints and internal events.

Greedy growth helps make the boundary approximation useful by absorbing compatible connections before fragments lock. In C–C≡N → C–C–N (Figure 2c), tolerance one can leave C–C separate from N. Exchanging the two carbons then changes the outer contribution from one weakening to a cleavage plus a formation. Tolerance two allows that connection inside one fragment, where its dependence is handled by the internal response. Greediness reduces such artificial boundaries but cannot guarantee independence: the ring–chain example still has a variable outer attachment. Equation (1) is consequently a model count, which may differ from the physical whole-reaction count.

The requested event window focuses decoding on low-event alternatives. Local bounds and actual branch witnesses first establish a best attained count U. When attained local minima compose, the smallest pre-sweep composition L has an attainable upper bound no greater than $L + 2$; only compositions that could improve the current best require further refinement. Unfinished local searches or incompatible local minima instead retain their lower bounds and an actual attained upper bound, without assuming that $L + 2$ is attainable. Branches whose lower bounds exceed $U + n$ are excluded. The studies below use a slack of $n = 5$.

The result retains branch identities, shared fragment responses, the event window and minimum/proof status. A later query can request a correspondence at a selected count or with specified atom pairs. It returns a concrete witness, established absence or an unresolved outcome under its allowance. Retaining a count interval does not assert that every count inside it is attainable. A proved model minimum, a best-found model count and a physical event minimum remain distinct.

2.7 Execution: parallel stages and shared computation

The stage boundaries also define parallel work. Reactions and directions are independent; seed/cut searches within a direction can run concurrently before their results are reduced together. Growth along one branch remains sequential because later fragments depend on earlier occupancy. For the reported batch workflow, matching and reduction finish before catalogue decoding begins. Different reaction/direction catalogues can then be decoded independently, with unique-fragment responses shared within each catalogue.

A persistent worker pool schedules these operations through the same interfaces used by direct local calls. Intermediate records can remain in memory or use aggregated storage. Compatible fragment proofs can be reused during growth, but each seed and cut keeps its own random decisions. Matching and decoding use a shared symmetry core, and bulk operations use native kernels. Scheduling and persistence do not determine which mappings are allowed. Detailed solver, cache and storage protocols, together with full pseudocode, appear in Appendix A.

3 Evaluation

3.1 Datasets and search settings

We evaluate all 1,851 Golden reactions (10) and the prepared collection of 47,974 EnzymeMap reactions (7). Reference labels are removed before search. Both studies use the prepared formal-bond-order matrices and explicit hydrogens; these results do not constitute a benchmark of continuously valued electronic-structure WBO inputs. EnzymeMap references are curated database mappings, so agreement is not independent validation of reaction mechanisms.

The common settings are discovery tolerance $\tau_{\text{iso}} = 2$, presence/cut threshold $b = 0.2$, hierarchical branch cap 100, random seed 42, both directions and the full single-edge sweep. Golden uses one and ten seed orderings per cut; EnzymeMap uses one. Decoding uses the original weights, thresholds 0.5/0.3, a five-second allowance per unique fragment and the best attained model count plus five. No reference mapping guides growth, pruning or starting-correspondence selection.

A separate paired Golden study evaluates current one-seed MAPPA and pinned SLAPMapper on identical prepared inputs and one CPU model. We distinguish standard binary SLAP, called on the uncut graph, from an expanded protocol combining binary and weighted modes, both directions and every single-edge cut. The latter is our exploration wrapper, not default SLAP or its published benchmark. Its comparison with default SLAP changes both search scope and weight mode; it is not a sweep-only ablation. Both methods retain their original algorithms, and external evaluation applies the same chemical endpoint-equivalence rule. The saved seed study and paired study have separately recorded runs and source identities; their recovery counts are not pooled.

3.2 Catalogue and decoded-reference verification

We report two external coverage measurements. *Catalogue recovery* asks whether the reference belongs to the retained matching relation. *Decoded recovery* requires an actual injective mapping from a saved starting correspondence and its joint symmetry choices, in a retained branch and event window. A positive certificate is replayed and its event count independently checked. A catalogue certificate may suggest a candidate, but cannot alone establish decoded membership.

The decoded audits preserve exact annotated heavy-atom indices and unmatched-heavy-atom requirements. Unannotated hydrogens must admit a feasible completion. Golden catalogue verification additionally permits endpoint chemical symmetries preserving charge, isotope, hydrogen count and stereochemistry; the Golden catalogue and decoded scores therefore differ in both representation and reference-equivalence policy. EnzymeMap uses exact indices for both. Recovery means inclusion in either search direction, not top-ranked prediction accuracy. Every reaction remains in the denominator, and unresolved queries remain separate from absence.

3.3 Timing and branch-event analysis

Mapping time is the sum of matching and reduction; decoding is reported separately. CPU seconds sum successful worker computation over both directions and all cuts, including stage input loading and result construction. Publication, external verification, historical failed attempts and parallel elapsed duration have separate accounting. Timings use the saved successful passes and their documented recoveries, rather than a newly measured uninterrupted run of one final performance revision. Exact source and measurement boundaries are recorded in the repository.

To locate a recovered reference within the decoded event landscape, let $\mathcal{C}_{\mathrm{GT}}$ contain the retained branch contexts that admit it inside the saved window. We measure

$$n_{\mathrm{GT}} = \min_{c \in \mathcal{C}_{\mathrm{GT}}} \min_{m \in c} E_c(m) - U, \tag{2}$$

where U is the overall detected count, minimized across both directions. Ground truth selects the qualifying contexts; their event minima are queried without fixing the reference assignments. Both terms use the same saved all-atom decoder model. This measures where a GT-containing branch begins, not the physical event count of the reference itself. Missing references and unresolved offsets remain outside numeric bins.

We separately count the distinct heavy-atom mapping classes actually attainable at the detected minimum, taking the union across retained branches and both directions. Chemical endpoint equivalence identifies repeated representations, and hydrogen permutations do not multiply the choice count; each class must admit a full valid completion. The ranking score remains the saved all-atom model count. If a reaction has K choices and its reference class is present, a uniform selector succeeds with probability $1/K$; otherwise it succeeds with probability zero. This defines expected random-choice accuracy without a trained selector. Incomplete counts contribute bounds, not exact values. A GT-containing branch's minimum need not be attained by GT itself, so this measurement is separate from Equation (2).

4 Results

4.1 Reference coverage on Golden and EnzymeMap

All requested matching, reduction and decoding outputs are available: 3,702 directions in each Golden setting and 95,948 in EnzymeMap. Table 2 and figure 3 separates catalogue membership from the independently audited decoded window.

Table 2 Reaction-level reference coverage. Percentages use every reaction. Decoded columns use exact annotated indices in the saved model min+5 window. Golden catalogue membership also allows endpoint chemical symmetry; EnzymeMap catalogue membership uses exact indices. Unknown queries are not counted as misses.

Study	Reactions	Catalogue	Decoded	Absent	Unknown
Golden, 1 seed	1,851	1,834 (99.082%)	1,832 (98.974%)	16	3
Golden, 10 seeds	1,851	1,843 (99.568%)	1,841 (99.460%)	8	2
EnzymeMap, 1 seed	47,974	47,798 (99.633%)	47,796 (99.629%)	178	0

One-seed Golden decoding confirms 1,832/1,851 references (98.974%), with 16 absent and three unresolved. Ten seeds confirm 1,841 (99.460%), with eight absent and two unresolved. Ten references become newly confirmed; one previously positive reaction remains unresolved in the ten-seed audit, giving a net gain of nine confirmed reactions and no proven loss. Catalogue recovery is 1,834 and 1,843, respectively. One ten-seed directional catalogue query exhausted its allowance, but its opposite direction establishes that reaction's catalogue recovery.

EnzymeMap decoding confirms 47,796/47,974 references (99.629%), compared with 47,798 (99.633%) in the catalogue. The two additional catalogue references have 22-event witnesses outside their saved 14–19 windows; independent decoded queries confirm their exclusion. All 94,820 positive directional decoded certificates

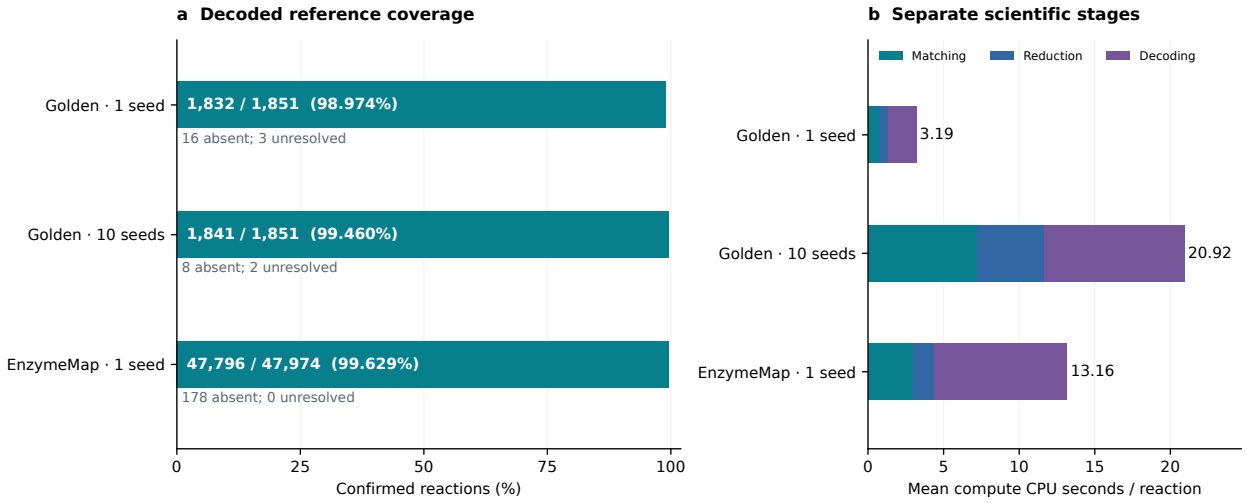


Figure 3 Reference coverage and computational cost. (a) Confirmed reference inclusion in the saved decoded min+5 window, with missing and unresolved reaction counts retained. This is inclusion among alternatives, not top-ranked accuracy. (b) Mean compute CPU per reaction, separated into matching, reduction and decoding. Both search directions contribute. Golden matching was reused for the corrected postprocessing passes; EnzymeMap combines original successful work with successful recoveries. External verification and publication are excluded.

passed replay and count checks, with no invalid certificate. Forty-five reactions unresolved by bounded decoded queries are conclusively absent from the unchanged containing catalogue, establishing their absence from its decoded subset. Their 250 unresolved directional queries remain recorded as such. The combined reaction accounting is therefore 47,796 recovered, 178 absent and zero unresolved.

4.2 Separate mapping and decoding costs

Table 3 and figure 3 report compute CPU rather than parallel latency. One-seed Golden requires 1.31 CPU seconds per reaction for matching plus reduction and 1.88 for decoding. Ten seeds require 11.64 and 9.28 seconds, respectively. The corresponding EnzymeMap means are 4.41 and 8.75 seconds, with medians of 2.15 and 3.46 seconds. The median combined EnzymeMap cost is 5.80 seconds; medians of separate stages need not add.

Table 3 Mean compute CPU seconds per reaction. Both directions and the full cut sweep are included. Matching plus reduction is the mapping stage. Decoding is separate; publication and external verification are excluded. These are recorded completed-stage costs, not a hardware-normalized comparison with the archived methods.

Study	Matching	Reduction	Mapping	Decoding	Total
Golden, 1 seed	0.77	0.54	1.31	1.88	3.19
Golden, 10 seeds	7.27	4.37	11.64	9.28	20.92
EnzymeMap, 1 seed	3.02	1.39	4.41	8.75	13.16

The distributions have substantial tails. Golden's maximum directional decoding cost is 52.69 seconds with one seed and 316.46 seconds with ten; its corresponding medians are 0.476 and 1.964 seconds. EnzymeMap's combined per-reaction computation has a 95th percentile of 41.86 seconds and a 99th percentile of 128.58 seconds. Including publication and measured worker overhead raises EnzymeMap's mean from 13.16 to 14.60 CPU seconds per reaction. Historical failed attempts contribute a further approximately 27.93 sampled CPU hours; external verification and diagnostic work are excluded from these algorithm costs.

4.3 Seed count and computational scaling

Increasing Golden from one to ten seeds increases matching CPU by 9.42-fold, reduction by 8.09-fold and decoding by 4.95-fold. Matching therefore scales approximately linearly with seed count in the measured one-to-ten-seed comparison. Decoding grows less because the extra searches produce a reduced set of shared fragments and branch compositions rather than ten independent decoding jobs. These measurements describe the cost of additional exploration and its coverage gain; they are not a controlled isolation of individual acceleration mechanisms.

4.4 Paired coverage and speed comparison with SLAP

The paired study recovers 1,839/1,851 references with MAPPA (99.352%), 1,794 with expanded SLAP (96.921%), and 1,620 with standard binary SLAP across both directions (87.520%; Figure 4). Expanded SLAP has 52 confirmed misses and five unresolved reactions; standard bidirectional SLAP has 230 misses and one unresolved. MAPPA has 12 misses and no unresolved catalogue queries. The independent comparator audit reconstructs its actual returned heavy mappings and repeats chemical-equivalence verification with a separate graph-matching implementation: all 3,683 completed expanded-SLAP direction verdicts agree.

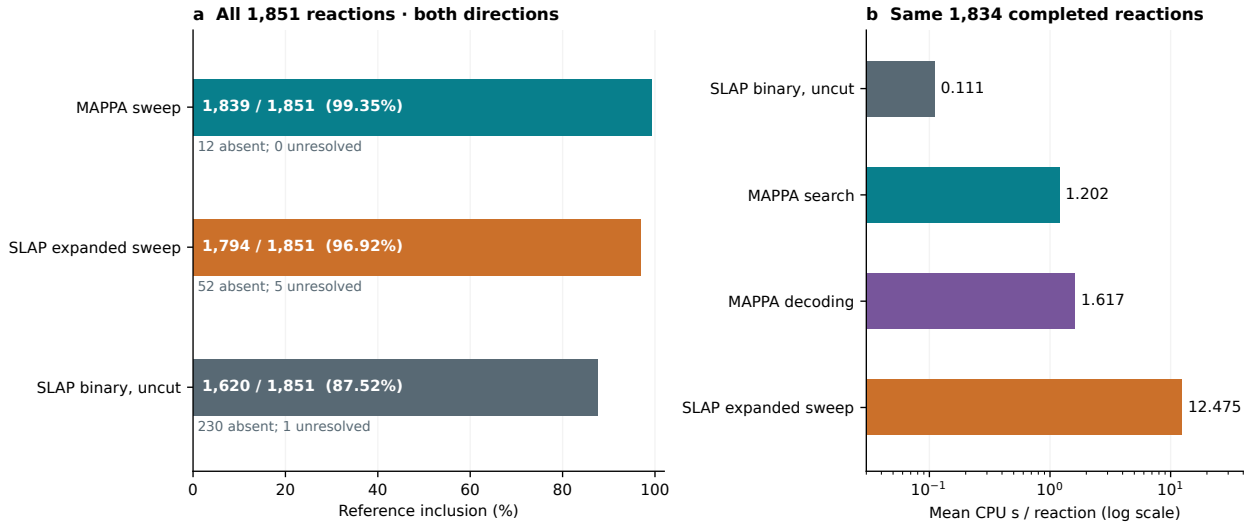


Figure 4 Coverage and cost depend on exploration scope. (a) Reference inclusion under identical chemical-equivalence evaluation over all 1,851 Golden reactions. Unknowns remain outside the recovered count. Standard SLAP is binary and uncut; expanded SLAP combines both weight modes and the cut sweep. All bars include both directions. (b) Mean compute CPU on the same 1,834 reactions completed by all protocols, with MAPPA decoding separate from search. The 17 excluded reactions remain in the coverage denominator. Timing is logarithmic and measures accumulated CPU, not parallel latency. The expanded wrapper raises SLAP coverage to approximately 97%, while increasing its computational cost.

Default forward-only SLAP recovers 1,591/1,851 (85.95%). This is close to its original paper's 86.9%, which also measures reference inclusion among alternatives but uses a 1,758-reaction Golden version (9). It is therefore an independent reproduction of similar behavior, not an exact replication of that dataset. Our archived first-returned default-SLAP score is 1,399/1,851 (75.58%). The retained alternatives tie at SLAP's lowest found cost; their output order is not a separate confidence ranking. In contrast, default single-answer interfaces such as RXNMapper and LocalMapper are evaluated directly on their selected prediction (Appendix C.1).

On the 1,834 reactions completed by all three protocols, mean search costs are 0.111 CPU seconds for standard binary SLAP, 1.202 for MAPPA, and 12.475 for expanded SLAP, summing both directions. MAPPA decoding adds 1.617 seconds. Thus MAPPA search is 10.38 times cheaper than the expanded SLAP sweep, but standard SLAP is 10.80 times cheaper than MAPPA search. The broader standard-SLAP/MAPPA pair contains 1,849 completed reactions and retains a slower standard-SLAP tail: their means are 0.487 and 1.259 seconds, a

2.58-fold difference. These are paired measurements on Intel Xeon Gold 6442Y processors, excluding loading, publication and external verification. Failed attempts remain separately accounted; excluded reactions are not treated as fast completions.

4.5 How many choices remain at the minimum?

The minimum-event rule produces a compact shortlist in the saved one-seed Golden study (Figure 5). Exact counts are established for 1,829 reactions; 22 remain incomplete under the counting allowance. Among the certified cases, 1,370 have one choice, 352 have two, 96 have three to five, and 11 have six or more. The certified-subset mean is 1.36 choices. Even allowing arbitrary larger counts for unfinished cases, the full-dataset median is one and its 95th percentile is three. We do not estimate the full-dataset mean from the completed subset.

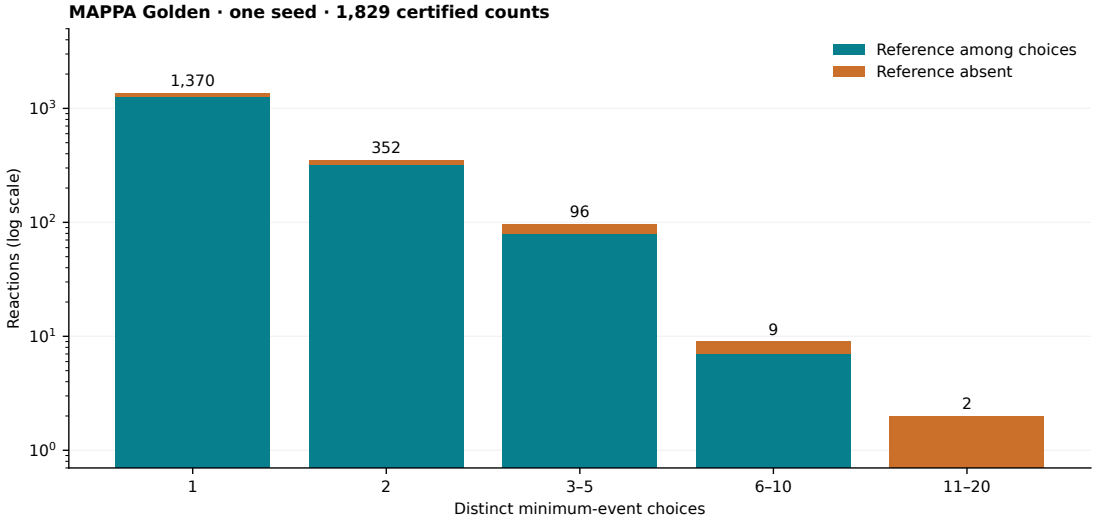


Figure 5 Distinct choices under minimum-event selection. Counts cover 1,829 of 1,851 one-seed Golden reactions with certified totals; 22 incomplete counts are excluded from the bars. Equivalent heavy-atom relabellings and hydrogen permutations count once. Colors indicate whether the reference is among the choices at the saved all-atom detected model minimum. The vertical axis is logarithmic. This is a count of attainable mapping classes, not branches or raw symmetry permutations.

The reference is confirmed at the detected minimum in 1,691 reactions, absent there in 158, and unresolved in two: minimum-set coverage is therefore 91.36–91.46%. Expected uniform-choice accuracy is 78.31–79.45%, with unfinished counts contributing explicit bounds. For example, Golden case 1694 has eight distinct minimum-event choices, including the reference, giving a random selector probability 1/8. These results quantify the decision left to a downstream selector. They do not establish top-1 superiority over another mapper, and a reference outside the minimum set would require selection from a wider event window.

4.6 Where reference-containing branches enter the event window

Figure 6 reports the preliminary distribution defined in Equation (2), with logarithmic reaction counts. Offsets are established for 45,737 reactions: 1,620 Golden and 44,085 EnzymeMap reactions have a GT-containing branch at the overall detected count; another 32 EnzymeMap reactions have offsets one through four. There are 3,891 pending offset checks, in addition to the separately reported absent and unresolved references.

Zero-offset certificates can establish the answer immediately, so the completed subset overrepresents zero. The higher-offset distribution remains incomplete and does not yet support full-dataset proportions. Both sides of

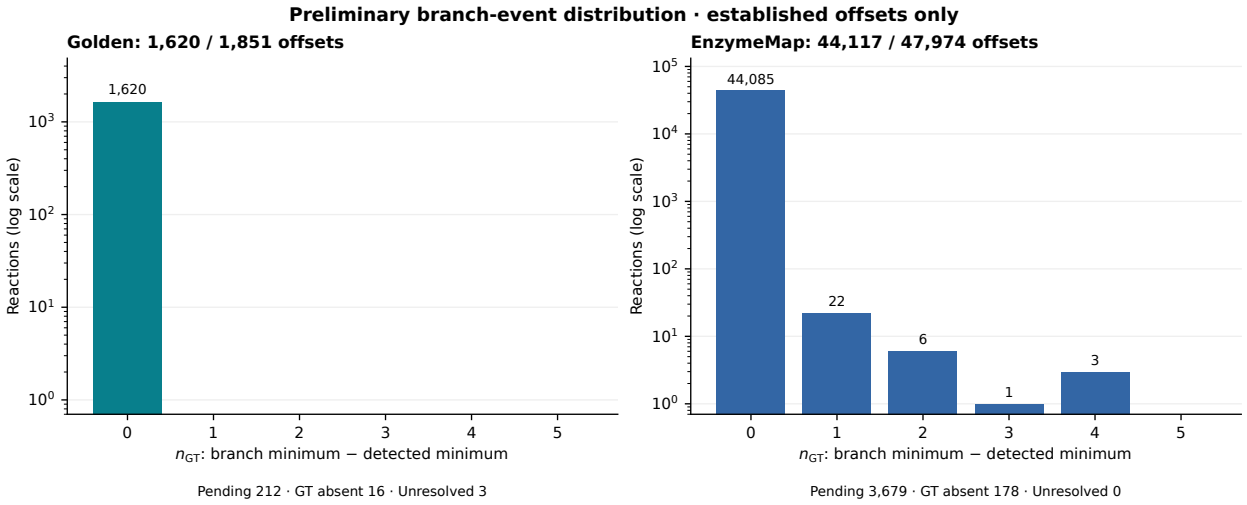


Figure 6 Preliminary location of GT-containing branches. The horizontal coordinate is the smallest event count of a retained GT-containing branch minus the overall detected count; the vertical axis is logarithmic. Ground truth selects the branch, rather than constraining the minimum witness within it. One-seed results use tolerance 2, cap 100 and saved min+5 responses. Pending checks and unresolved memberships are excluded from bars and labeled explicitly. Empty bins must not be interpreted as final zero frequencies.

the subtraction use the same decoder count, including hydrogen and the ordinary-boundary approximation. The offset measures the reported model landscape, not physical event-bin accuracy.

4.7 Chemical interpretation of retained alternatives

A correspondence describes net atom origins and bond changes without specifying intermediates or elementary steps. Figure 7 illustrates why retaining alternatives can matter. In Golden case 9, the annotated assignment places the alcohol oxygen in the product ring and the ketone oxygen in water. A second assignment exchanges those oxygen origins, while a third also changes two carbon assignments. The second has fewer events than the reference; the first and third have equal event totals but different correspondences. Neither a minimum event count nor a single total identifies the annotated transformation.

The figure uses selected archived witnesses as a chemical illustration. It is not an exhaustive output of the current decoder, and the alternatives are not validated reaction pathways. The historical coordinate/WBO study in Appendix C similarly examines event alternatives under unverified mappings; it is not a mapping-accuracy benchmark.

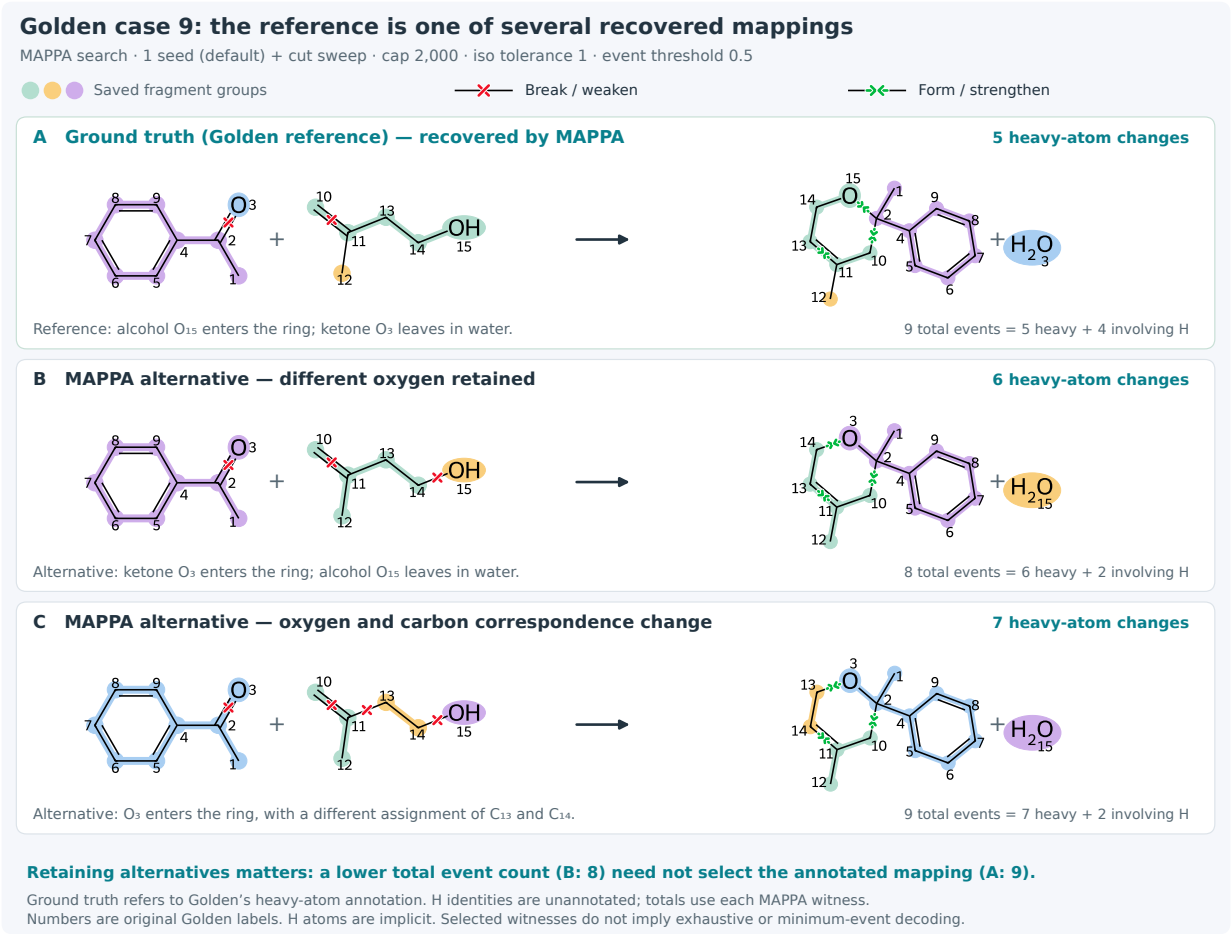


Figure 7 A reference correspondence and two alternatives for Golden case 9. A matches all 15 annotated heavy-atom pairs. B exchanges the oxygen origins; C also exchanges C_{13}/C_{14}. Colors identify matched fragments. Black lines show endpoint bonds, red × marks show cleavage or weakening, and green inward arrows show formation or strengthening. H atoms are implicit in the drawing but included in event totals; each witness minimizes H-involving changes for its fixed heavy-atom correspondence. "Ground truth" denotes the dataset annotation. These archived witnesses were obtained without reference labels using one-seed sweep search, cap 2,000 and $\tau_{\text{iso}} = 1$; events use threshold 0.5.

5 Discussion

MAPPA separates the existence question needed during growth from the compressed alternatives needed at saturation, and then separates correspondence discovery from event interpretation. Atom-set reduction gives repeated fragments one shared identity. The decoder reuses local symmetry-event responses while retaining correlated occupancy and the special dependence introduced by a swept bond. These module boundaries make search, representation and interpretation independently inspectable and reusable.

The benchmarks demonstrate high reference inclusion with complete saved mapping and decoding outputs, while preserving the distinction between catalogue and decoded coverage. More Golden seeds recover additional references at increased computation, with matching cost approximately proportional to seed count over the measured range. The paired SLAP comparison shows the trade-off directly: default SLAP is cheaper, whereas MAPPA explores a broader alternative set and costs less than the expanded SLAP sweep. Returning alternatives is shared by both methods; the comparison concerns their coverage, organization and cost.

Several limits remain scientifically important. Greedy trajectories, sampled seeds, single-edge cuts and hierarchical caps restrict discovery. Selecting one original fragment record per atom-set pair and one decoded

starting orbit can omit alternatives present in discarded discovery records. Bounded improvement can retain an unproved minimum. None of these restrictions establishes exhaustive correspondence recovery.

The fixed ordinary-boundary model is also an approximation. In the EnzymeMap audit, 2,801 positive directional witnesses have different model and physical whole-reaction event counts; the ten-seed Golden audit records 32 such witnesses. Both counts are retained. Model-window membership is therefore not physical event-bin accuracy, and a low model count alone does not identify the annotated or most plausible chemistry. The branch-offset figure assesses organization within that model, not a replacement accuracy measure.

Retained alternatives can support reaction-path preparation: atom-index assignment affects endpoint interpolation (3), and different permitted correspondences could supply starting structures for nudged elastic band calculations (8). A separate chemical selector could turn the same catalogue into a single-answer prediction, using minimum event count as a surrogate and reaction-specific preferences to resolve ties or retain a higher-event candidate. These are downstream opportunities, not demonstrated improvements in ranking or transition-state prediction. The current Golden and EnzymeMap studies also do not validate continuously valued WBO inputs or every stereochemical attribute. The archived comparator, seed/cap, direction and coordinate analyses remain in the supplement with their original protocols; only the separately paired current comparison supports its stated speed ratios.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden and EnzymeMap are distributed by their original authors (10; 7).

Acknowledgements

We gratefully acknowledge the longstanding contributions of the Matter Lab's current and past group members (matter.toronto.edu), and in particular those of the El Agente team (elagente.ca). C.C. acknowledges support from the DOE grant with the University of Minnesota, award #A006801504, and the Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education (RS-2025-02634334). Y.K. was supported by the CIFAR AI Safety Catalyst Award (Catalyst Fund Project #CF26-AI-001). J.B.P.S. acknowledges funding of this project by the Natural Sciences and Engineering Research Council of Canada (NSERC) Alliance Grant #ALLRP587593-23 (Quantamole). A.W. acknowledges support from the Canada Graduate Scholarships—Doctoral (CGS-D) program.
This research is part of the University of Toronto's Acceleration Consortium, which receives funding from the CFREF-2022-00042 Canada First Research Excellence Fund. Computations were performed on the Niagara supercomputer at the SciNet HPC Consortium. SciNet is funded by the Canada Foundation for Innovation; the Government of Ontario; the Ontario Research Fund—Research Excellence; and the University of Toronto.
A.A.-G. thanks Anders G. Frøseth for his generous support and acknowledges the generous support of NSERC and the Canada 150 Research Chairs program. A.A.-G. and V.B. also acknowledge the generous support of Natural Resources Canada and the Canada 150 Research Chairs program and the University of Toronto's Acceleration Consortium, which receives funding from the CFREF-2022-00042 Canada First Research Excellence Fund.
This work was supported by the Defense Advanced Research Projects Agency (DARPA) under Agreement No. HR0011262E022. This work was also supported by the AI2050 program of Schmidt Sciences.
Molecular graphics and analyses were performed with UCSF ChimeraX (6; 12), developed by the Resource for Biocomputing, Visualization, and Informatics at the University of California, San Francisco, with support from National Institutes of Health R01-GM129325 and the Office of Cyber Infrastructure and Computational Biology, National Institute of Allergy and Infectious Diseases.
We thank Kjell Jorner and Robert Pollice for helpful discussions.

References

[1] Shuan Chen, Sunggi An, Ramil Babazade, and Yousung Jung. Precise atom-to-atom mapping for organic reactions via human-in-the-loop machine learning. *Nature Communications*, 15:2250, 2024. doi: 10.1038/s41467-024-46364-y.

[2] Chython contributors. Chython: molecular and reaction processing toolkit. https://github.com/chython/chython. Evaluated version 2.18 with chython-rxnmap 2.0; model hash archived.

[3] Deb Sankar De, Marco Krummenacher, Bastian Schaefer, and Stefan Goedecker. Finding reaction pathways with optimal atomic index mappings. *Physical Review Letters*, 123:206102, 2019. doi: 10.1103/PhysRevLett.123.206102.

[4] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340. Springer, 2008. doi: 10.1007/978-3-540-78800-3_24.

[5] EPAM Systems. Indigo cheminformatics toolkit. https://github.com/epam/Indigo. Evaluated version 1.46.0.

[6] Thomas D. Goddard, Conrad C. Huang, Elaine C. Meng, Eric F. Pettersen, Gregory S. Couch, John H. Morris, and Thomas E. Ferrin. UCSF ChimeraX: Meeting modern challenges in visualization and analysis. *Protein Science*, 27:14–25, 2018. doi: 10.1002/pro.3235.

[7] Esther Heid, Daniel Probst, William H. Green, and Georg K. H. Madsen. EnzymeMap: curation, validation and data-driven prediction of enzymatic reactions. *Chemical Science*, 2023. doi: 10.1039/D3SC02048G. https://doi.org/10.1039/D3SC02048G.

[8] Graeme Henkelman, Blas P. Uberuaga, and Hannes Jónsson. A climbing image nudged elastic band method for finding saddle points and minimum energy paths. *The Journal of Chemical Physics*, 113(22):9901–9904, 2000. doi: 10.1063/1.1329672.

[9] Shin-ichi Koda and Shinji Saito. General and scalable atom-to-atom mapping via Weisfeiler–Lehman-like approximate graph matching. *ChemRxiv*, 2025. doi: 10.26434/chemrxiv-2025-hthwn. Preprint, version 1, 25 November 2025.

[10] Arkadii Lin, Natalia Dyubankova, Timur I. Madzhidov, Ramil I. Nugmanov, Jonas Verhoeven, Timur R. Gimadiev, Valentina A. Afonina, Zarina Ibragimova, Assima Rakhimbekova, Pavel Sidorov, Andrei Gedich, Rail Suleymanov, Ravil Mukhametgaleev, Joerg Wegner, Hugo Ceulemans, and Alexandre Varnek. Atom-to-atom mapping: A benchmarking study of popular mapping algorithms and consensus strategies. *Molecular Informatics*, 41(4):2100138, 2022. doi: 10.1002/minf.202100138.

[11] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.

[12] Eric F. Pettersen, Thomas D. Goddard, Conrad C. Huang, et al. UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Science*, 30:70–82, 2021. doi: 10.1002/pro.3943.

[13] S. A. Rahman, G. Torrance, L. Baldacci, S. Martínez Cuesta, F. Fenninger, N. Gopal, S. Choudhary, J. W. May, G. L. Holliday, C. Steinbeck, and J. M. Thornton. Reaction decoder tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13):2065–2066, 2016. doi: 10.1093/bioinformatics/btw096.

[14] Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Advances*, 7(15):eabe4166, 2021. doi: 10.1126/sciadv.abe4166.

A Technical algorithm details

The complete stage sequence is summarized in Algorithm A1. The following sections give the constraints and implementation details underlying the main-text description.

Algorithm A1 Matching, reduction and separate decoding

Require: Endpoint graphs, discovery policies, seed/cut plan, branch cap and event window
for all requested search directions **do**
  for all cut conditions and seed orderings **do**
    Initialize the conditional search hierarchy
    while a retained continuation has an eligible fragment seed **do**
      Grow using one-witness existence queries
      At saturation, construct distinct compressed placements
      Continue each placement with earlier assignments locked
      Prune and recount the hierarchy from leaves to root
    end while
    Retain terminal compositions and compact provenance
  end for
  Share fragments by atom-set pair; reduce repeated branch compositions
end for
Complete the matching/reduction stage before batch decoding
for all reaction/direction catalogues **do**
  Inventory unique fragments and prepare their shared symmetry-event responses
  Compose branch contributions and the sweep-endpoint correction
  Retain queryable branches within the requested event window
end for

A.1 Exact existence queries during growth

An existence query contains the proposed source atoms, their accepted preservation edges and completed-fragment locks. Element-compatible destination domains exclude targets already occupied by other fragments. Each accepted edge supplies a binary support relation satisfying target bond presence and the discovery tolerance. Source nonedges and unselected molecular edges do not add constraints.

More formally, let V_R, V_P denote endpoint atom sets, Z_R, Z_P their elements and W_R, W_P their original bond-weight matrices. A partial correspondence $m : D \subseteq V_R \to V_P$ is injective. For every accepted preservation edge $(u, v) \in A$, its assignment satisfies

$$Z_R(u) = Z_P(m(u)), \quad W_P(m(u), m(v)) \geq b, \quad |W_R(u, v) - W_P(m(u), m(v))| \leq \tau_{\text{iso}}. \tag{3}$$

Here b is the graph-presence threshold and τ_{iso} is the discovery tolerance. These constraints apply to the selected growth graph, not every bond in the completed atom set.

Propagation removes occupied singleton destinations and destinations without support in a neighbor's domain. An empty domain or incompatible singletons establish infeasibility. A bipartite distinct-target check can reject domains that cannot accommodate an injective assignment. These are necessary conditions; surviving ambiguous cases require search. The solver chooses a constrained atom, propagates its tentative destination and stops at the first complete witness. Proving absence can require exploring every remaining feasible choice. Positive and negative answers use the same constraints.

Direct witness extension and indexed fragment reuse precede this query. They establish existence only. Neither supplies a different continuation policy nor converts a timeout into absence. Completed-fragment assignments remain fixed while assignments internal to the current growing fragment may change.

The molecular overview in Figure 1 uses constructed alcohol placements in panel a and methyl acetate saponification in panels b and c. Hydrogens participate in matching but are implicit in the drawings. Exchanging the acetate oxygen images adds two internal changes to two boundary changes at event threshold 0.5. Black lines show endpoint bonds. These illustrations are not a complete search trace, and their formal-bond-order signatures are not distinct mechanisms or resonance-normalized transformations.

A.2 Completion, local symmetry and global placement

Completion searches the finished fragment constraints for distinct target atom-set placements. It excludes already represented sets or certified global placement orbits within the search. It does not enumerate all internal assignments before deduplicating them. Target sites that cease to be globally equivalent remain separate placements, even when their local matched subgraphs share symmetry.

For each local fragment, nauty receives an element-colored binary graph with individually fixed anchors. Its compact generators retain correlated moves of rings, arms and their attached atoms. Continuous weights do not label this graph; they remain in discovery and event predicates. Permitted local rearrangements are compositions of these binary generators. An accepted discovery mapping can be followed by a source and a target local action without reimposing discovery tolerance on the rearranged result. The event response evaluates the resulting original-weight changes.

The global action used to compress later continuations is more restrictive. It must preserve the full target weight matrix, active connectivity, current locks and every source-to-target compatibility predicate relevant to subsequent growth. Such an action moves the whole later continuation, preserving occupied targets and its matching problem. A local fragment action cannot be substituted for this global equivalence proof. Saved placement operations retain their order; unrestricted closure of an ordered sequence would permit combinations not represented by the search.

A.3 Hierarchical caps and provenance

Each retained placement creates a child continuation. After a completed step, count surviving continuations bottom-up. At an over-limit node, discard its largest child subtree, recount and repeat; ties retain the earlier decision. An ancestor then sees the reduced count. The completed-fragment limit separately rejects a placement expansion exceeding its allowance. Neither operation counts internal symmetry assignments as separate branches, and both retain explicit cap evidence.

A shared search state may have several incoming histories. The retained graph keeps parent links and terminal references rather than unfolding every path. States and payloads with no surviving descendant are released. Returned histories and cut provenance remain available for inspection, but are not independent fragment-decoding jobs. A branch cap bounds retained choices, not every temporary allocation or total memory.

A.4 Fragment identities and composition reduction

Reduction reuses the fragment inventory in each search graph and unifies it across seeds and cuts. A completed fragment's key is its source/target atom-set pair within a fixed endpoint problem and policy. Keep the first original record intact. Other starting witnesses, growth routes, domains or anchors do not create additional identities and are not unioned into broader permissions. Raw archives retain the original decisions; the reduced catalogue retains the selected record and compact provenance. When reducing an explicitly selected path subset, the representative must come from that subset.

Native projection converts retained states into shared fragment identifiers. Composition keys use those identifiers together with the ordered placement context. Independent decisions can commute; overlapping local or global choices retain their order. Equal keys share one family record and aggregate cut/partition provenance. Reduction uses structural equality, not symmetry solving or event queries. The atom-set key must not be confused with a permission to map its atoms arbitrarily.

A.5 Shared event core and bounded starting-map improvement

Each unique fragment is prepared once. For a selected starting correspondence, the shared core stores binary symmetry generators, fixed event contributions and sparse variable original-weight terms. It separates operation components only when no variable event or requested endpoint joins them. A generator moving two arms couples those moves even if each arm also has its own independent internal exchanges. Event-preserving operations can share computation only when they also preserve all endpoints relevant to the query. No complete group or Cartesian table of mappings is stored.

Starting-map improvement first compares the saved witness with lower bounds. One bound matches pair-weight multisets within element-pair types while relaxing atom consistency. Another matches incident bond assignments and then atoms injectively; each event touches two atoms, so half the incident cost, rounded up, is a lower bound. The stronger bound is retained. These relaxations can rule out improvement but need not be attainable.

Optimize the selected fragment's binary symmetries. Search additional discovery correspondences only when its recorded growth connectivity differs from final internal connectivity on either endpoint and the current witness remains unsettled. New starting correspondences must satisfy the saved discovery domains, anchors and accepted edges; subsequent binary symmetry choices are not rejected by those discovery constraints. Preparation, correspondence improvement and symmetry optimization share a five-second per-fragment deadline. Attaining a valid bound stops work immediately. Otherwise retain an actual best witness and incomplete status. An orbit minimum does not certify a minimum over discarded starting orbits.

Branch assembly binds local response operations to their shared branch choices. Final partition pieces own disjoint internal contributions. Weight-preserving global placements move endpoints while retaining those local costs. Ordinary boundaries remain fixed under the declared model. The sweep correction follows jointly reachable endpoint pairs through the saved action order, preserving dependence on variable internal events. When internal costs are constant, this is an endpoint-pair calculation, not a complete symmetry-assignment enumeration.

For a requested event interval, the saved response searches its retained choices and returns a concrete mapping with its ordered choices, absence after complete exclusion, or unknown under a limit. Event counts and mappings can be replayed independently. Preparing a queryable response does not assert that every count in its numerical interval is attainable. Physical whole-reaction event counts can differ from the independent-boundary model and are recorded separately during external audits.

A.6 Indexed fragment reuse and execution

The reaction-local pool stores one original fragment per atom-set pair. A restricted saved witness proves a growth proposal when it covers the requested source atoms, uses available target atoms, respects locks and supports every required edge at the current tolerance. Atom-set containment alone is insufficient when occupancy or edge requirements have changed. An inverted index intersects atom, edge-support and fixed-assignment sets to find a valid proof. A miss invokes the ordinary existence query; no negative cache is used.

Growth continues with its own random proposals. Cut/trial streams are distinct and reproducible, and completion chooses its representatives independently of the temporary existence witness. The pool therefore does not replay a previous trajectory. Its accounted retention limit bounds cached records and index storage, not total stage memory. Returned search graphs expose their unique-fragment inventories directly to reduction.

Persistent execution shares compatible reaction snapshots at task boundaries. A permanent first snapshot and replaceable latest snapshot support concurrent readers; a validated first snapshot can be used when the latest stored version cannot be loaded. This is recovery of reusable data, not an alternative scientific algorithm. A single publication owner atomically replaces complete versions, and stale publishers cannot overwrite newer ones. Stored snapshot count is bounded per reaction rather than per cut or seed.

Separate stage APIs return typed matching, catalogue and decoding records. In-memory and aggregated file-backed transport preserve shared immutable arrays and fragment relations. Worker pools schedule those operations without changing their scope. For the reported batch protocol, a global matching/reduction barrier precedes decoding. Recovery retains completed outputs and consumed budgets; resource errors remain distinguishable from mapping absence. The main manuscript reports successful compute costs separately from failed attempts and external evaluation.

B Reproducibility and measurement scope

The repository's `manuscript/REPRODUCIBILITY.md` links source pins, immutable input identities, benchmark records and the evidence used to generate tables and figures. The public Python API documents direct matching, reduction, decoding and conditional queries. Figure building runs no searches. Publication records omit execution-site identifiers.

The main-text Golden measurements combine the saved one- and ten-seed matching passes with corrected completed reduction and decoding passes. They are not new uninterrupted end-to-end measurements. The matching passes predate indexed containment; the paper treats their seed scaling as an empirical workload comparison, not a cache ablation. EnzymeMap combines original successful tasks and successful recoveries with unchanged scientific settings. Performance corrections preserved the mapping relation and event model. CPU totals include computation, stage loading and record construction; publication, external audits and sampled earlier failures are separate. CPU models were not retained in the exported full-campaign summaries, so no hardware-normalized ranking follows.

Every accepted decoded-reference witness is checked for replay, injectivity, annotated assignments and an independently counted model total inside the saved window. Its physical total is stored separately. The Golden decoded audit uses exact indices; its catalogue audit permits endpoint chemical equivalence. EnzymeMap uses exact indices for both. A conclusive absence from the unchanged containing catalogue implies absence from the decoded subset; that inference is recorded separately from a direct negative decoded query. Bounded unknown queries are never relabeled as direct negatives.

The branch-offset snapshot contains one row per reaction. Its 45,737 exact offsets, 3,891 pending offsets, 194 absent references and three unresolved memberships sum to 49,825. Easy zero-offset certificates are resolved first, so the partial histogram is not an unbiased estimate of the final distribution. Both compared counts use the saved all-atom decoder model. No heavy-only reference count enters the subtraction.

The supplementary September studies below retain their original settings, evaluators and comparison cohorts. Their archived paired search timings use an Apple M4 Pro CPU; the cap comparison uses an AMD EPYC 9J14. Search, publication and event-query boundaries differ across those studies and are stated with each table. They are not pooled with the current main-text timings. The WBO-generation method for the archived coordinate inputs is not recorded in the available metadata, so regeneration of those matrices from molecular structures is not established.

C Archived September evaluations

The following studies used their recorded September implementations. They retain the original numerical results and comparison scopes; they are not reruns of the current algorithm or direct performance comparisons with Section 3.1.

C.1 Archived datasets and comparator protocol

Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. These chemical colors define reference equivalence; they do not imply that element/WBO matching itself enforces every chemical attribute. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (1). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (9). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; "SLAP sweep" denotes this extension, not a published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (14; 2; 5; 13).

Table A1 Archived evaluation configurations. Both use $b = 0.2$, $\tau_{iso} = 1$, explicit H, and uncut plus single-edge searches. One seed per cut and the sweep define the default setting. Archived Golden measurements cover sweep search; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multi-component structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. The archived study compared the lowest event count found and the patterns achieving that count. MAPPA used supplied WBO matrices; SLAP used XYZ-derived binary graphs. Returned SLAP heavy-atom assignments received hydrogen refinement and were scored on the same WBO matrices as MAPPA. The comparison covers those saved witnesses, not all possible SLAP hydrogen assignments or a fresh current-implementation evaluation.

Released comparators received the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid endpoint chemistry, duplicate atom labels and other failures remained in the denominator. Binary SLAP was retained as the stronger default configuration; bidirectional comparisons used both binary and weighted modes. SLAP returns multiple candidates, whereas the other comparators return at most one explicit bijection. These results describe the evaluated interfaces and scoring rule, including its difference from RDT's chemical-change criterion. The repository records released models, versions, adapter settings and failure frequencies.

C.2 Archived evaluation configurations

Table A1 lists the archived settings. Search did not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix B).

C.3 Archived Golden reference recovery and search cost

In the archived campaign, the one-seed MAPPA configuration recovered 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table A2 and figure A1). These scores belong to that campaign, separate from Section 3.1. The paired comparison includes 43 references recovered only by MAPPA and 5 only by SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional MAPPA seeds yielded modest recovery gains at increasing search cost; two seeds did not improve recovery over one.

The paired archived controls attribute 345 additional reference recoveries to the MAPPA sweep and 135 to the SLAP sweep. Without the sweep, one-seed MAPPA had lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.3); symmetry expansion alone cannot restore discarded placements. These differences are not measurements of a current no-sweep control.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. In the archived comparison, combining both directions added 31 recoveries over smaller-first alone, demonstrating complementary alternatives (Table A3).

In these archived measurements, one-seed MAPPA had a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately doubled the mean cost without improving recovery. These Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets; neither establishes a current-implementation speed ranking.

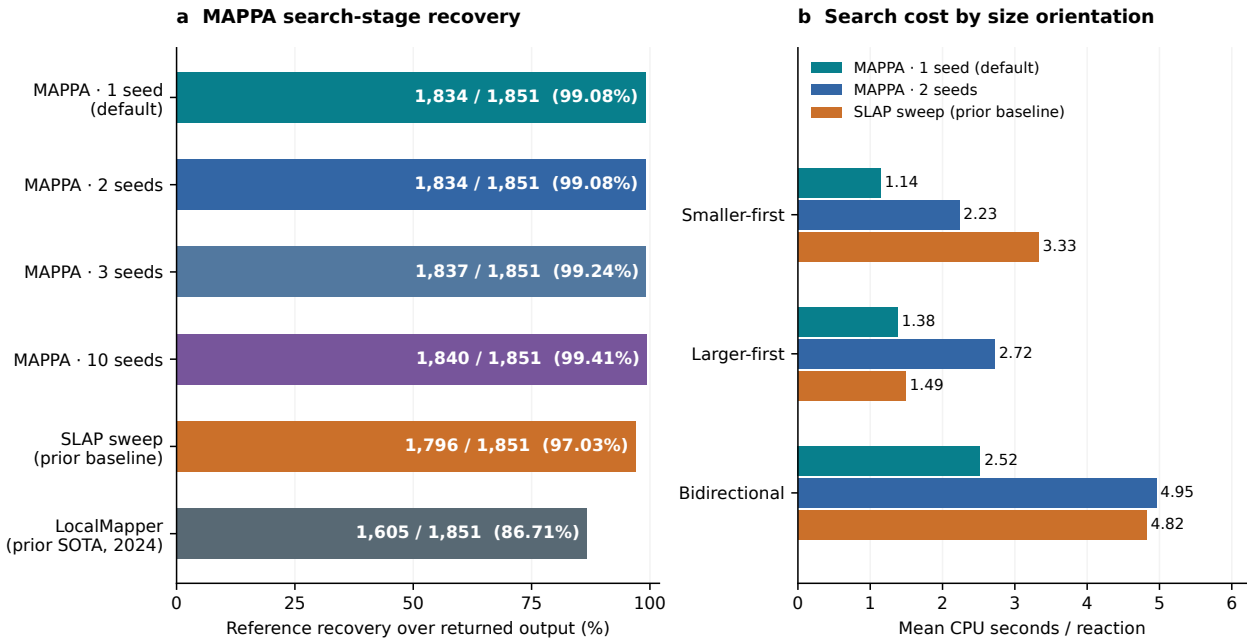


Figure A1 Archived Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. MAPPA and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use the archived evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in Tables A2 and A3.

In the archived cap study, increasing the branch cap from 100 to 2,000 yielded no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A8). The uncut search gained 19 confirmed recoveries. Thus, in that study a larger branch budget did not substitute for changing the growth conditions.

Table A2 Golden reference recovery and runtime in the archived evaluation. Recovery uses all 1,851 primary-evaluation records. MAPPA retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. MAPPA uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
MAPPA	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
MAPPA	2 seeds, sweep	Families	1,834 (99.08%)	1.619
MAPPA	3 seeds, sweep	Families	1,837 (99.24%)	2.265
MAPPA	10 seeds, sweep	Families	1,840 (99.41%)	6.993
MAPPA	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075[a]
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056[a]
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294[a]
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612[a]
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184[a]
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021[a]

Table A3 Archived recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first reverses the source and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in Appendix B.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
MAPPA, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
MAPPA, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

C.4 Archived bond-event comparison on 140 coordinate reactions

The archived coordinate study used one-seed MAPPA sweep search with a branch cap of 2,000 and returned complete mappings for all 140 reactions. At cap 100, cases 123 and 125 returned no complete mapping. The 2,000-branch configuration therefore supplies this comparison, while 100 remains the library default. These completion counts and the event-window results below have not been remeasured with the current implementation.

The archived MAPPA output matched or lowered the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table A4 and figure A2). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remained absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table A4 Archived minimum bond-event comparison on 140 reactions. MAPPA uses one-seed sweep search at cap 2,000. The first three columns count reactions by MAPPA's minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by MAPPA among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	MAPPA fewer	Equal	MAPPA more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above MAPPA's own minima, the archived study queried through one event above the higher of the two SLAP minima for each reaction. MAPPA retained 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at MAPPA's own minima. Every saved family was exhausted or certified within its window; this establishes completeness for those retained families, not exhaustive structural search.

Table A5 Archived stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. MAPPA uses one-seed sweep search at cap 2,000. These stage costs exclude MAPPA decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix B.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table A5 cover different outputs: MAPPA search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

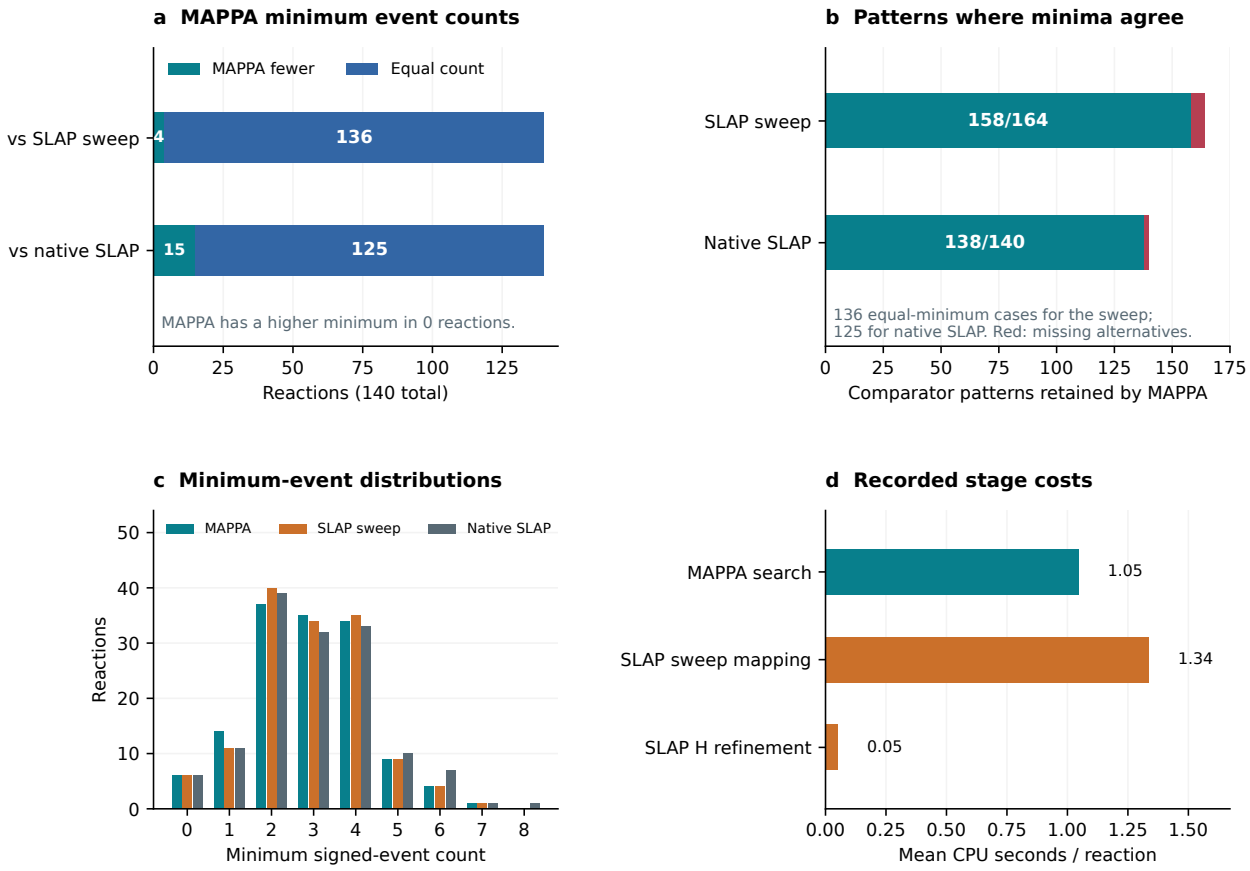


Figure A2 Archived bond-event comparison on 140 coordinate reactions. MAPPA uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal MAPPA events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by MAPPA at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding MAPPA decoding (Table A5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

C.5 Archived event-window query scope

For each coordinate reaction, the archived event-window limit was one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families were exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness. This explicit whole-reaction query is separate from ordinary decoding into shared local fragment responses.

Table A6 Archived complete event-window queries on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

The query used event invariance to certify constant families and encoded remaining choices in Z3 (4). A family was certified by exhaustion, a valid lower bound excluding the window, or unsatisfiability after excluding all discovered classes. An exhausted resource budget remained incomplete. The explicit query adapter can simplify adjacent nested action groups by exact generator-membership checks: if $H \leq G$, then $HG = GH =$

G. This adapter simplification is separate from ordinary catalogue reduction in Appendix A.4. Completeness within an event window concerns the saved families, not exhaustive structural search.

C.6 Archived Golden nonrecoveries

Table A7 The eleven archived ten-seed Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms. This is not the nonrecovery set of the current one-seed validation.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The archived uncapped traces below demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

Two archived traces show how greedy extension excluded a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Growth preserved bonds that these correspondences require separating. Uncapped trajectories still missed both references, while reference-constrained fragment choices established their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposed the reference to matching; the single-edge sweep did not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries or traces of the current matcher. For the other five correspondence misses, the audit did not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

C.7 Archived branch-cap and CPU-time ablation on Golden

The archived study ran the twelve configurations in Table A8 on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the archived primary evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the archived

Table A8 Archived Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default setting.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
MAPPA uncut	100	1	1,489 (80.44)	0	0.040	0.085
MAPPA sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
MAPPA sweep	100	2	1,834 (99.08)	0	1.619	7.260
MAPPA sweep	100	3	1,837 (99.24)	0	2.265	10.556
MAPPA sweep	100	10	1,840 (99.41)	0	6.993	33.211
MAPPA uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
MAPPA sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
MAPPA sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
MAPPA sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
MAPPA sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

primary evaluation: two previously recovered cases are unresolved under this study's limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

C.8 Archived end-to-end timing on coordinate reactions

The archived September implementation completed search and event-window decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A9).

The historical study measured one-seed, cap-2,000 cut-sweep search followed by its event-window decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A9 Archived MAPPA search plus event-window queries. CPU seconds per completed reaction in the 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470